

CREDIT CARD FRAUD DETECTION

1. Introduction

Fraud detection in the case of credit cards is a key problem for many financial organizations. The fraudulent transaction may lead to substantial financial losses. Machine learning algorithms are utilized to detect the frauds in the case of credit card transactions.

2. Objective

The main objective of this project is to build a machine learning model that can accurately classify transactions as genuine or fraudulent.

3. Dataset Description

Dataset Name: Credit Card Fraud Detection Dataset

Features:

- Time
- Amount
- V1 to V28
- Class

Target Variable:

- Class = 0 (Genuine)
 - Class = 1 (Fraud)
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4. Tools and Technologies Used

- Python
 - Jupyter Notebook
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Scikit-Learn
 - VS Code
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5. Methodology

Step 1: Data Collection

The dataset was downloaded from Kaggle.

Step 2: Data Preprocessing

- Loaded dataset
- Checked missing values
- Removed duplicates
- Selected features and target variable

Step 3: Data Visualization

- Fraud vs Genuine Transaction Distribution
- Data Analysis using Graphs

Step 4: Model Training

Random Forest Classifier was used to train the model.

Step 5: Prediction

The trained model was used to predict fraudulent transactions.

Step 6: Evaluation

Model performance was evaluated using:

- Accuracy Score
- Classification Report
- Confusion Matrix

6. Results

The model achieved high accuracy in classifying fraudulent transactions.

Performance Metrics:

- Accuracy Score
 - Precision
 - Recall
 - F1 Score
-

7. Screenshots

Insert:

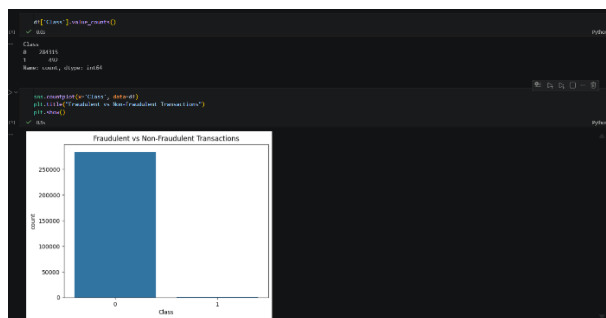
1. Dataset Screenshot

```
df = pd.read_csv("../dataset/creditcard.csv")
df.head()
```

	Time	V1	V2	V3	V4	V5	V6	V7	V8	V9	...
0	0.0	-1.359807	-0.072781	2.536347	1.378155	-0.338321	0.462388	0.239599	0.098698	0.363787	...
1	0.0	1.191857	0.266151	0.166480	0.448154	0.060018	-0.082361	-0.078803	0.085102	-0.255425	...
2	1.0	-1.358354	-1.340163	1.773209	0.379780	-0.503198	1.800499	0.791461	0.247676	-1.514654	...
3	1.0	-0.966272	-0.185226	1.792993	-0.863291	-0.010309	1.247203	0.237609	0.377436	-1.387024	...
4	2.0	-1.158233	0.877737	1.548718	0.403034	-0.407193	0.095921	0.592941	-0.270533	0.817739	...

5 rows × 31 columns

2. Class Distribution Graph



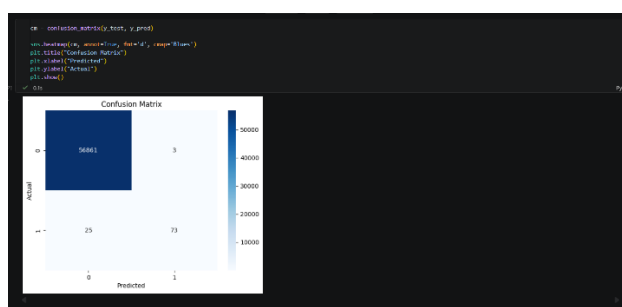
3. Accuracy Output

4. Classification Report

```
print("Accuracy:", accuracy_score(y_test, y_pred))
print(classification_report(y_test, y_pred))
```

Accuracy:	0.9998844229732
	precision recall f1-score support
0	1.00 1.00 1.00 56861
1	0.96 0.74 0.84 79
accuracy	0.96 0.97 1.00 56940
macro avg	0.98 0.87 0.92 56940
weighted avg	1.00 1.00 1.00 56940

5. Confusion Matrix



8. Conclusion

The project successfully demonstrates the use of machine learning techniques for detecting fraudulent credit card transactions. The developed model can help financial institutions identify suspicious activities and reduce fraud-related losses.

9. Future Enhancements

- Use Deep Learning models
 - Real-time Fraud Detection
 - Deployment using Streamlit
 - Integration with Banking Systems
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10. References

1. Kaggle Credit Card Fraud Detection Dataset
2. Scikit-Learn Documentation
3. Python Documentation